

App Presentation

PERFIT

Professor David Atienza
Professor Giovanni Ansaloni

Supervisor Rafa Medina

Christopher Julien Stocker (266575),

Yuanyuan Zheng (282304) ,

Vladimir Reymond (272260)

Semester Project
Fall 2022

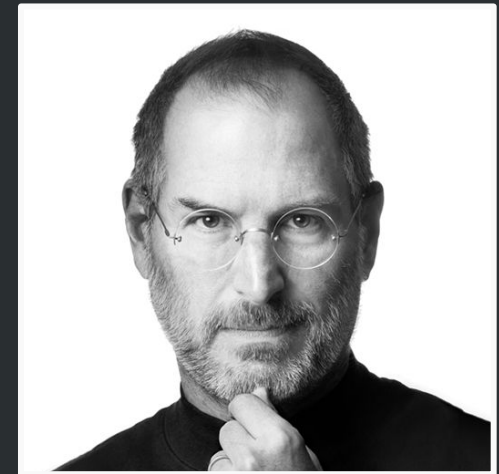
EPFL



App Idea

Decision Fatigue

refers to the deteriorating quality of decisions made by an individual after a long session of decision making.



Barack Obama, Steve Jobs, and Mark Zuckerberg have been known to reduce their everyday clothing down to one or two outfits in order to limit the number of decisions they make in a day. [wiki: Decision Fatigue]

App Idea

What if you could manage to everyday wear the perfect outfit without it being the same outfit?

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Perfit

To develop an Android application that alleviates the mental energy of outfit selection in the morning.

- The user builds a virtual closet and the application formulates an outfit suggestion.
- Suggestions are based on the user's desire for "Sports, Fashionable, Comfortable",
- Suggestions take into account clothes' cleanliness
- Suggestions take into account weather

Three Levels

Login & Registration

Dashboard Navigation

User Interaction

Firestore Structure



User

Firestore Authentication

- Email identifier
- User id

Verification done on FirestoreClass.kt

Constants Logged_In_Username → Stored on shared preferences

- avoid signing in at every launch
- reduce firestore reads

Firebase Structure - Firestore Storage

Clothes

- Database of all users' clothing items
 - Image: item's image link

Firestore Database

Top

- Availability
- Category
- Cleanliness
- Id Clothing
- Id User
- Image
- Purpose
- Season

Bottom

Shoes

User

- email
- name
- gender
- Tel

Outfit

- Id Outfit
- Id of 3 clothes
- Id User
- 3 Image URIs
- Purpose
- Season

1. Login



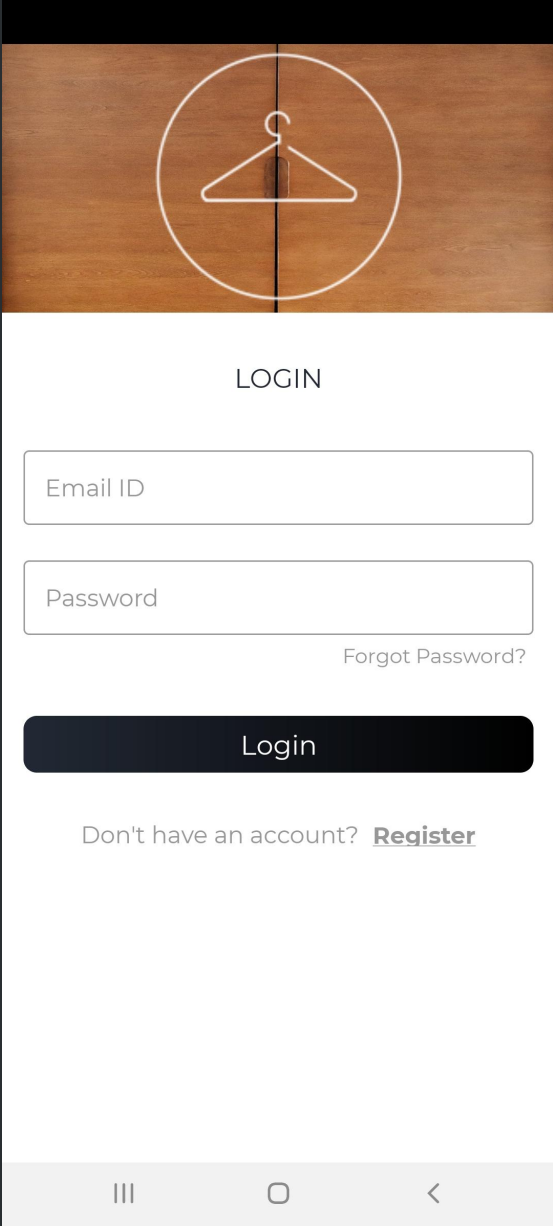
1. Login

LOGIN - Already registered?

REGISTER

FORGOT PASSWORD

COMPLETE USER INFORMATION



The image shows a mobile application interface for a login screen. At the top, there is a header image featuring a wooden background with a white circular outline and a hanger icon inside. Below the header, the word "LOGIN" is centered. There are two input fields: "Email ID" and "Password". To the right of the "Password" field is a link that says "Forgot Password?". Below the input fields is a dark blue button with the text "Login". At the bottom of the form area, there is a link that says "Don't have an account? [Register](#)". The bottom of the screen shows a standard Android navigation bar with three icons: a square, a circle, and a triangle.

LOGIN

Email ID

Password

[Forgot Password?](#)

Login

Don't have an account? [Register](#)

1. Login

LOGIN

REGISTER

FORGOT PASSWORD

COMPLETE USER INFORMATION

Mobile app UI for "CREATE AN ACCOUNT" screen. The screen displays a back arrow, a title "CREATE AN ACCOUNT", and five input fields for "First name", "Last name", "Email ID", "Password", and "Confirm Password". Below the input fields is a checkbox labeled "I Agree To The [Terms & Condition](#)". A "Register" button is located below the checkbox. At the bottom, there is a link "Already have an account? [Login](#)". The bottom of the screen shows a mobile navigation bar with three icons: a list icon, a home icon, and a back arrow.

1. Login

LOGIN

REGISTER

FORGOT PASSWORD

COMPLETE USER INFORMATION

<

CREATE AN ACCOUNT

First name

clothes

Last name

lover

Email ID

clothes.lover@gmail.com

Password

.....

Confirm Password

.....

☒

I Agree To The [Terms & Condition](#)

Register

Already have an account? [Login](#)

|||

○

<

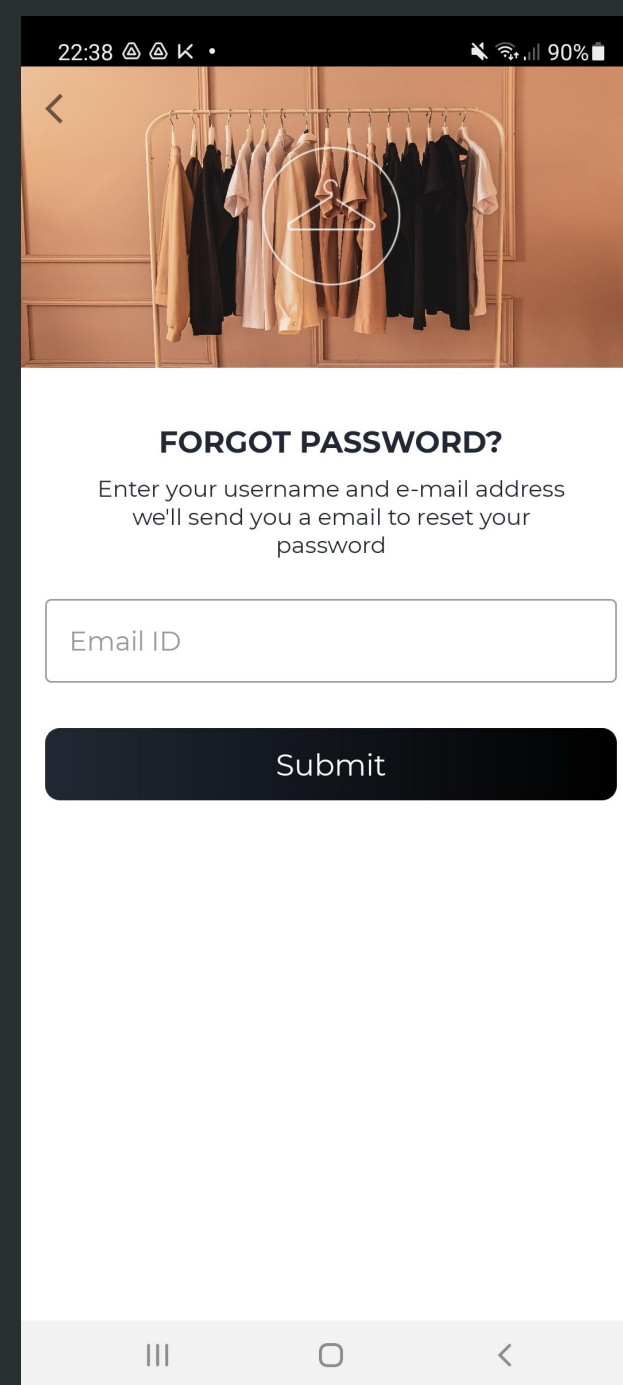
1. Login

LOGIN

REGISTER

FORGOT PASSWORD

COMPLETE USER INFORMATION



1. Login

LOGIN

REGISTER

FORGOT PASSWORD

COMPLETE USER INFORMATION

23:52 86%

COMPLETE PROFILE



First name
clothes

Last name
lover

Email ID
clothes.lover@gmail.com

Mobile Number
123456

☐ MALE ☒ FEMALE

☐ Save

2. Bottom Navigation

The user can go switch between three principle navigation menus.



Closet



Dashboard



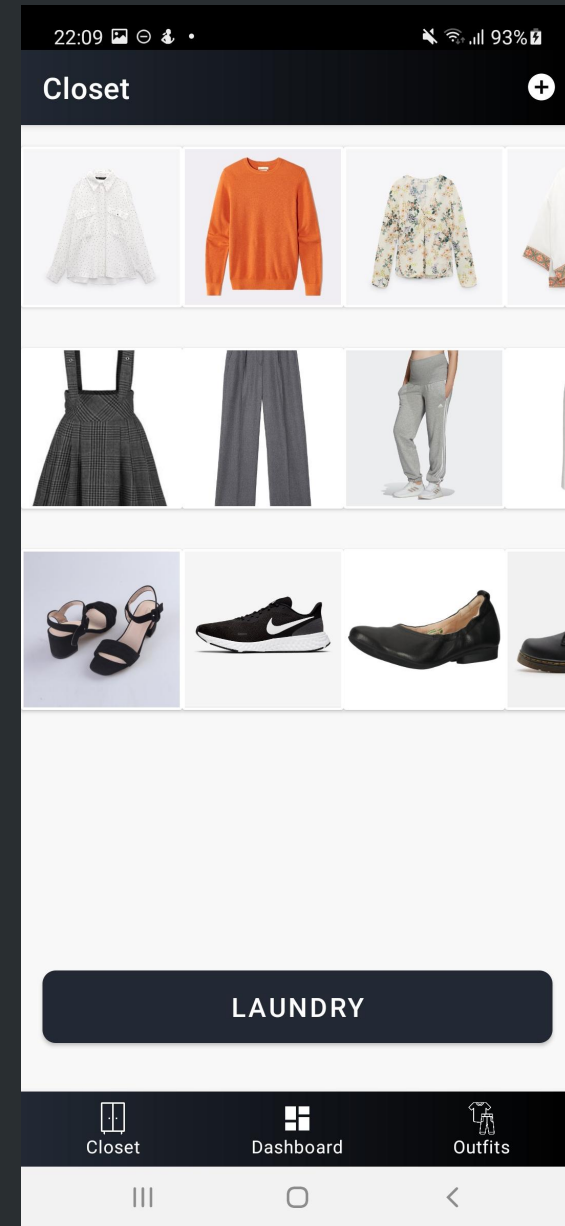
Outfits

2. Bottom Navigation

CLOSET

DASHBOARD

OUTFIT

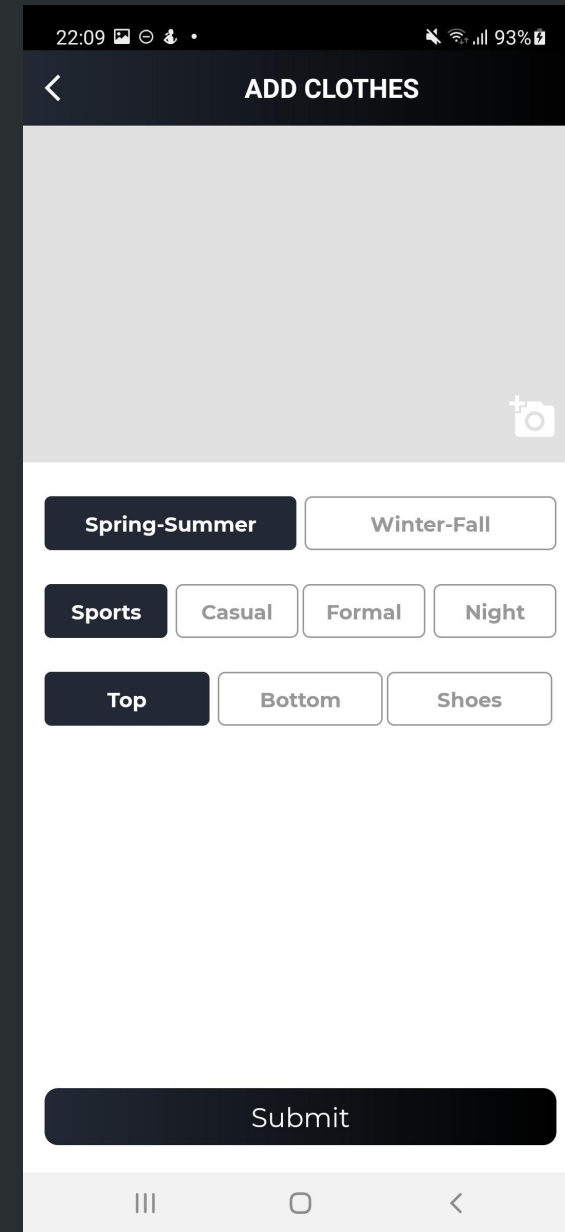


2. Bottom Navigation

CLOSET → Add clothes

DASHBOARD

OUTFIT

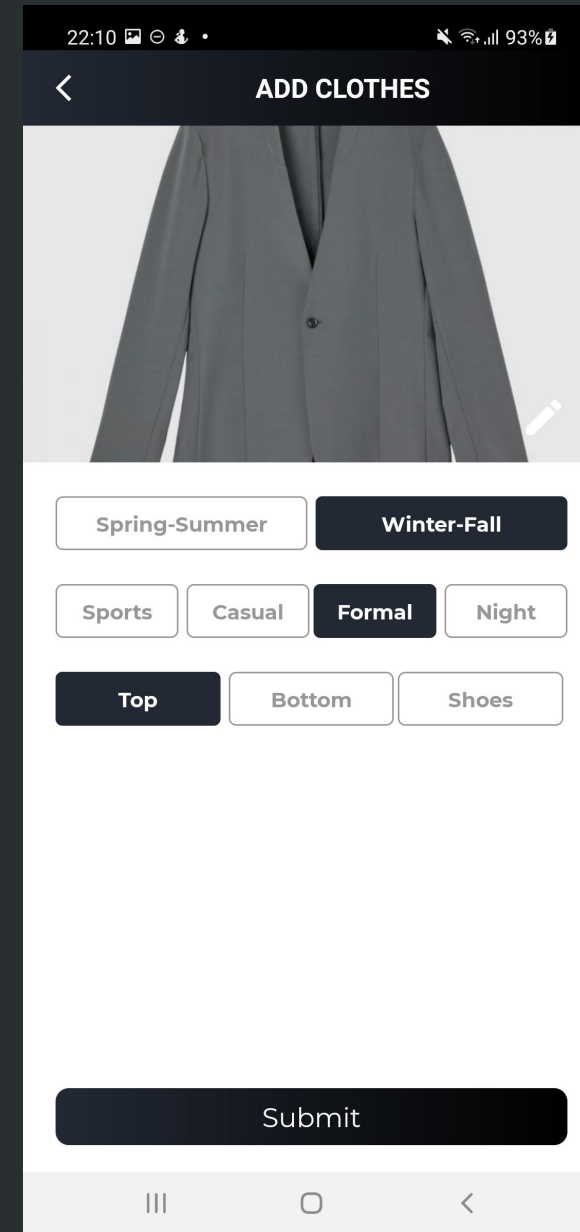


2. Bottom Navigation

CLOSET → Add clothes

DASHBOARD

OUTFIT

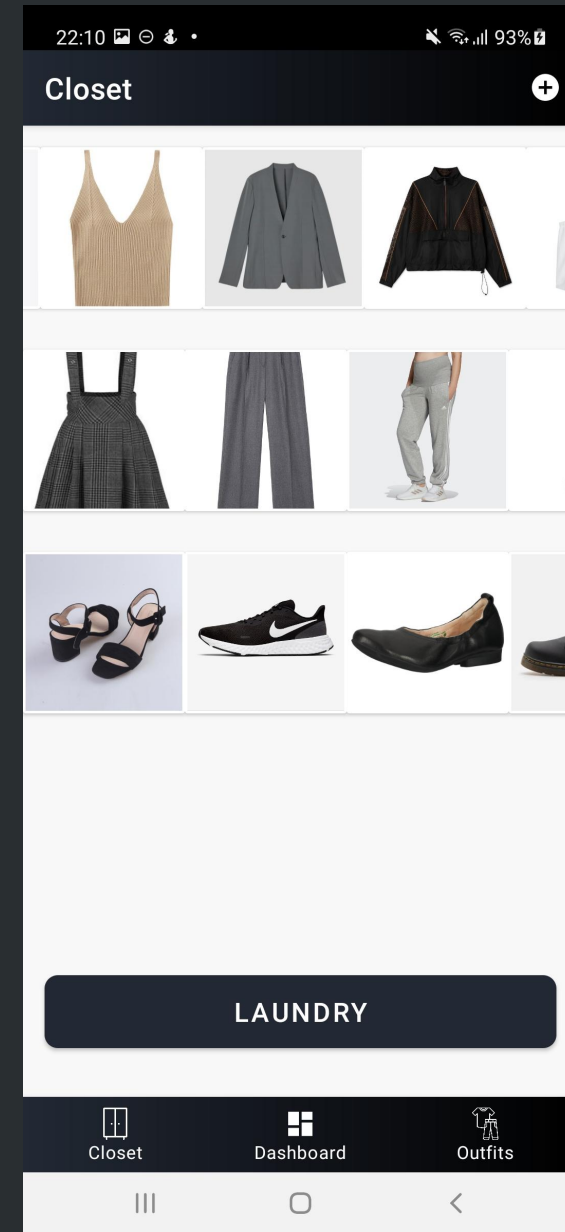


2. Bottom Navigation

CLOSET → Add clothes

DASHBOARD

OUTFIT

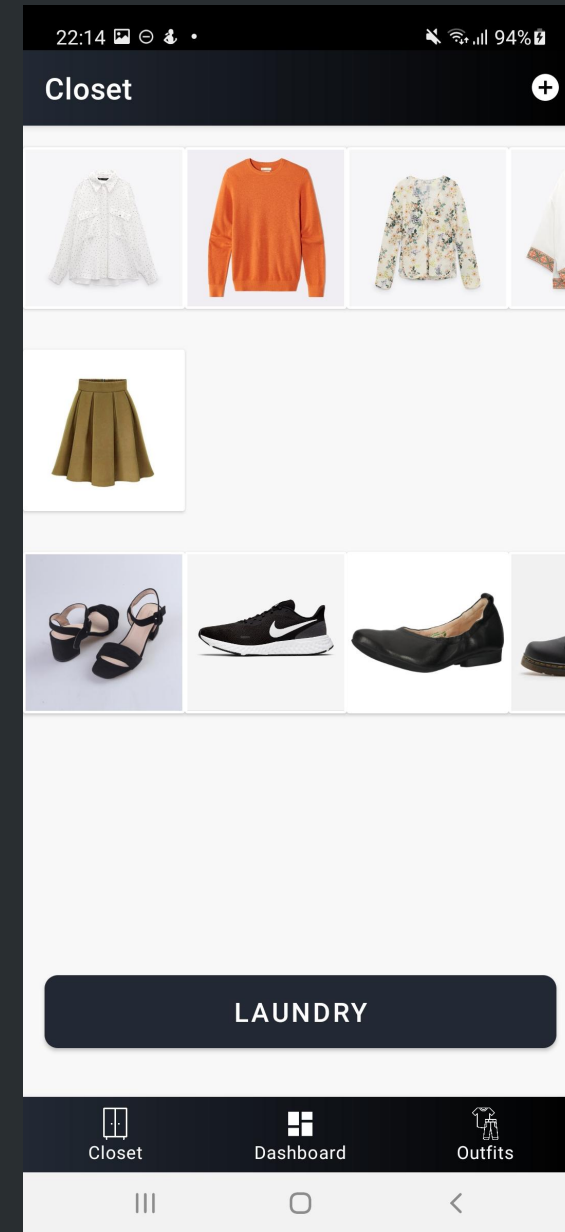


2. Bottom Navigation

CLOSET → Do laundry

DASHBOARD

OUTFIT

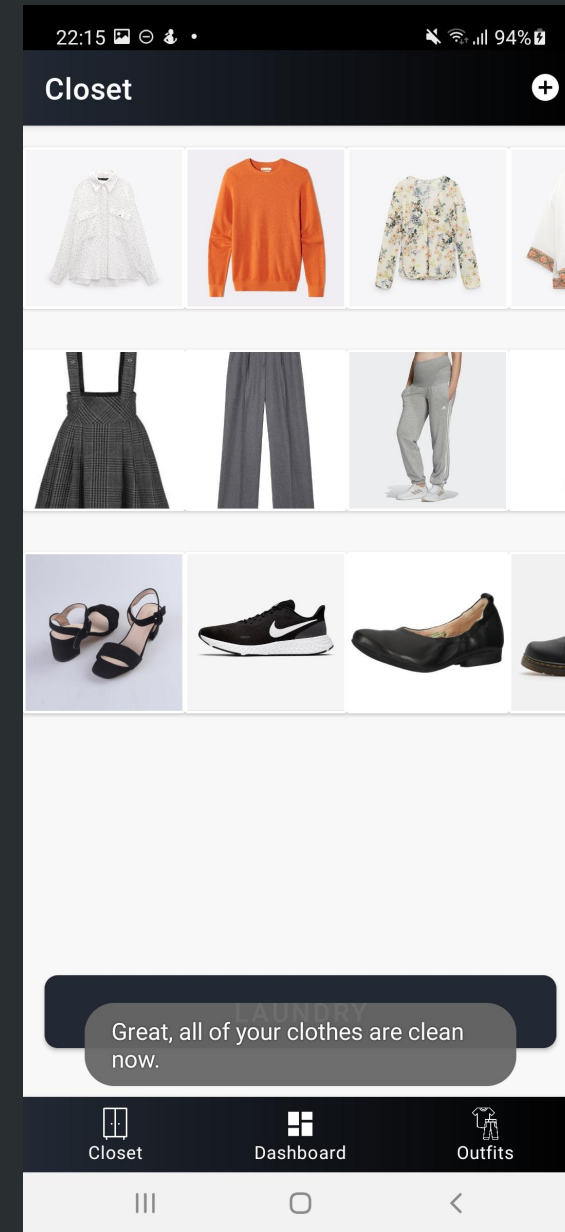


2. Bottom Navigation

CLOSET → Do laundry

DASHBOARD

OUTFIT

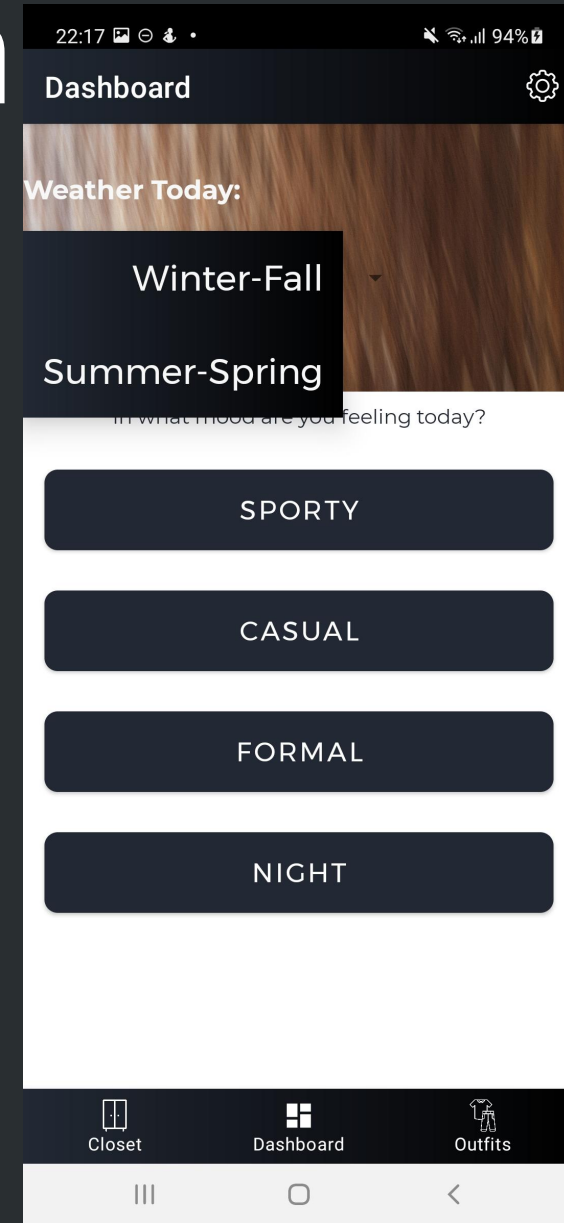


2. Bottom Navigation

CLOSET

DASHBOARD

OUTFIT

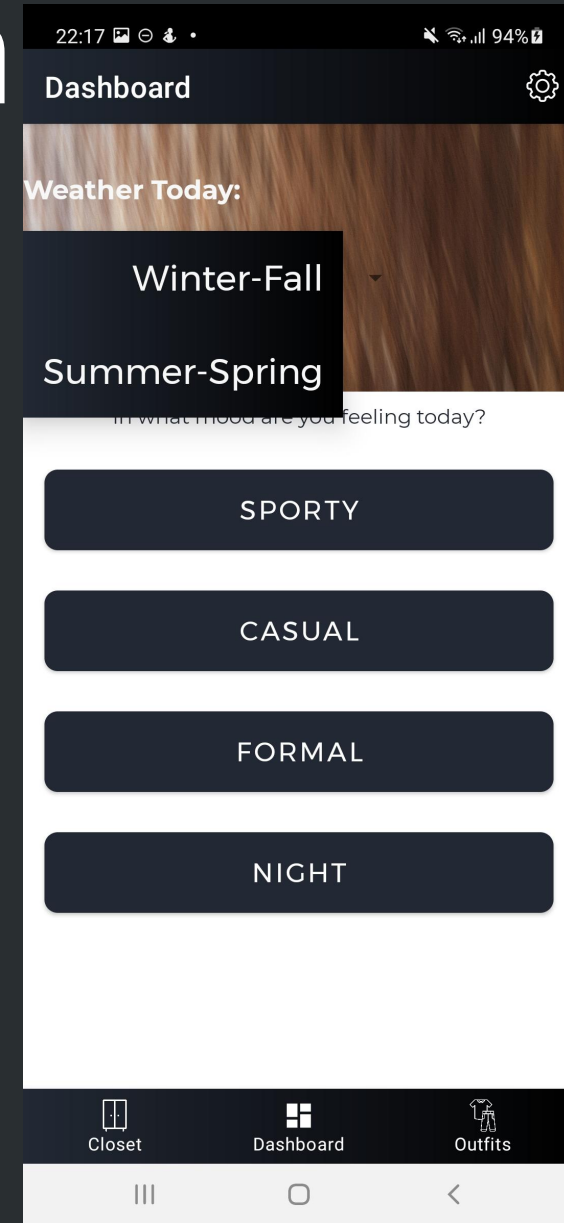


2. Bottom Navigation

CLOSET

DASHBOARD → outfit selection criterias

OUTFIT

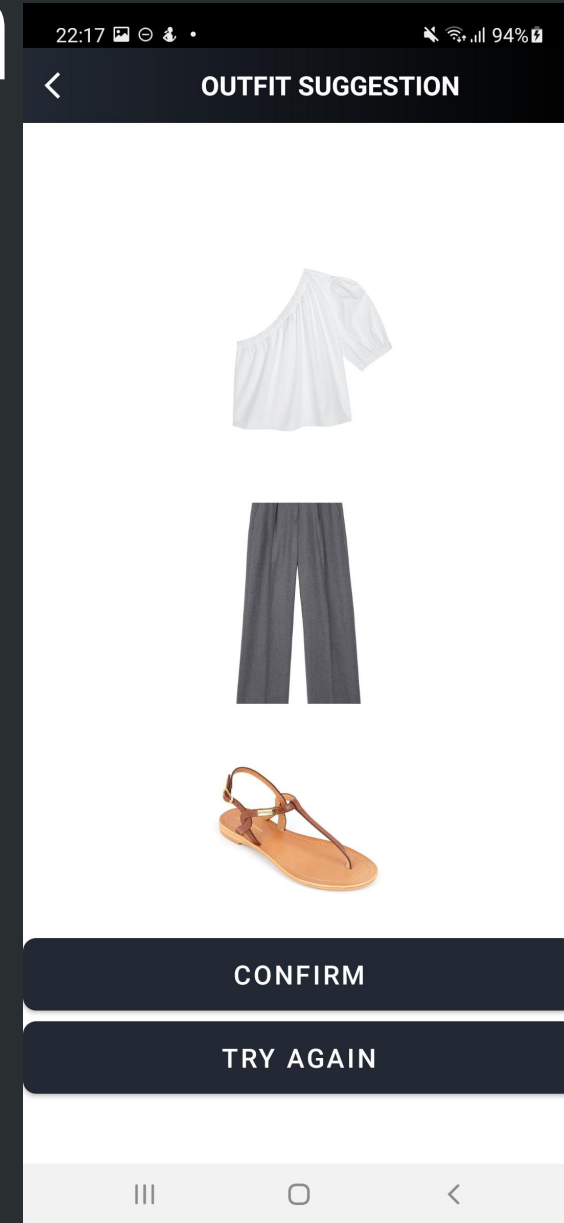


2. Bottom Navigation

CLOSET

DASHBOARD → daily outfit suggestion

OUTFIT

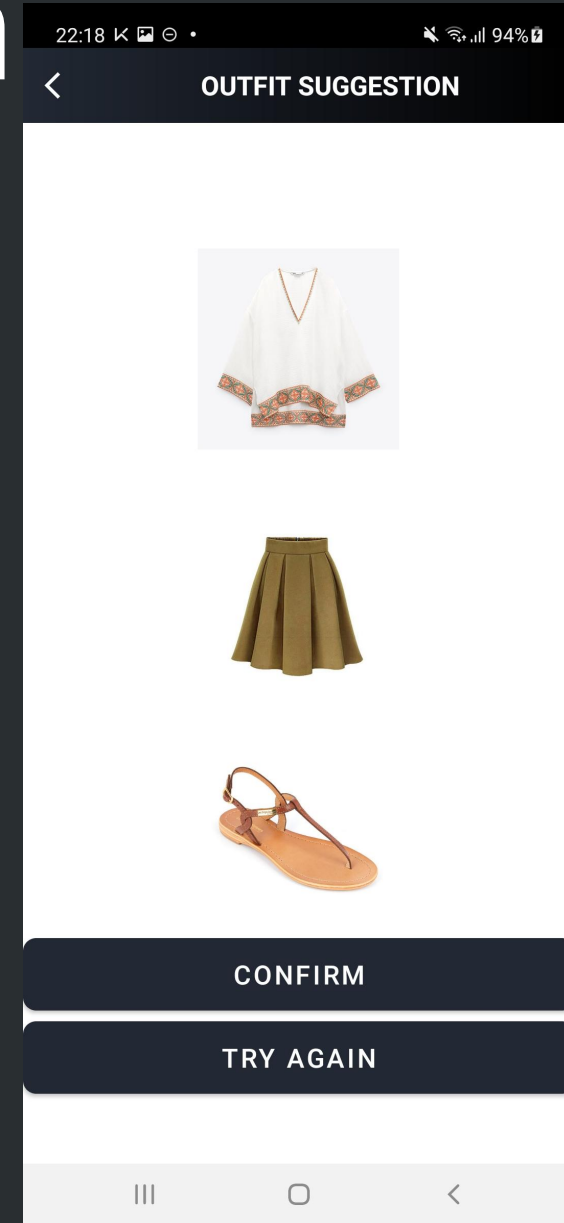


2. Bottom Navigation

CLOSET

DASHBOARD → discover more possible combinations

OUTFIT

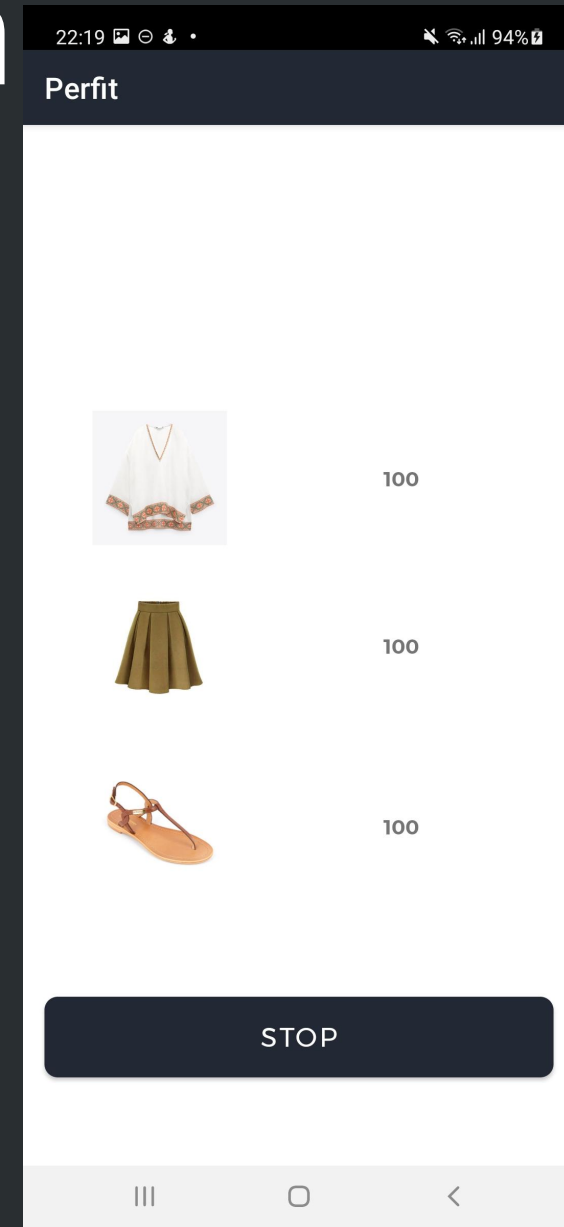


2. Bottom Navigation

CLOSET

DASHBOARD → decide your outfit of the day!

OUTFIT



2. Bottom Navigation

CLOSET

DASHBOARD

OUTFIT → outfits sorted by season and purpose

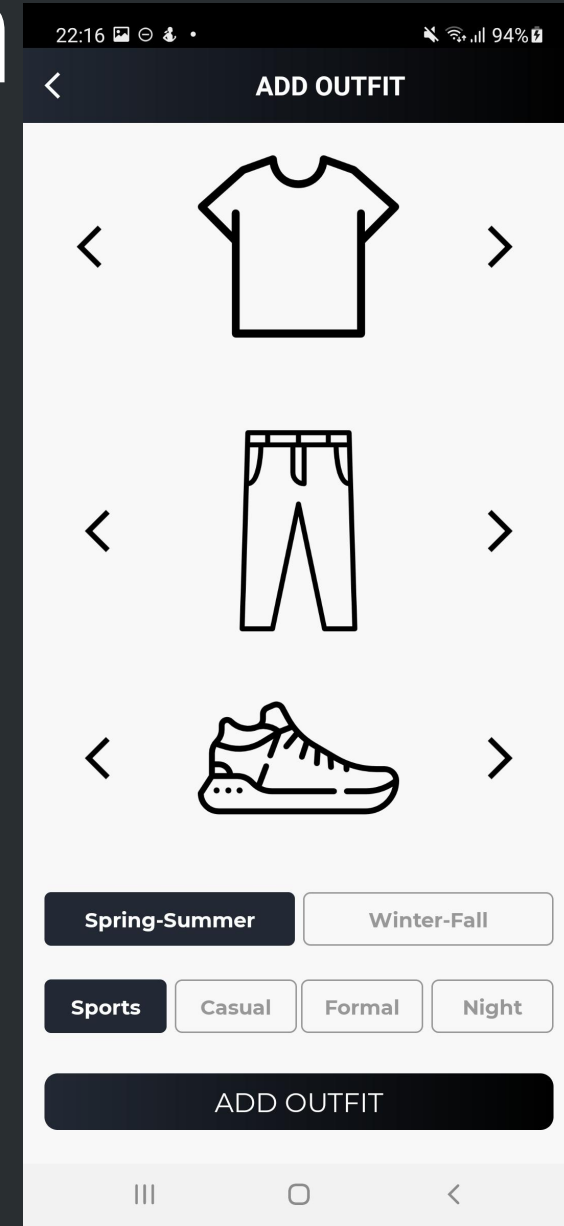


2. Bottom Navigation

CLOSET

DASHBOARD

OUTFIT → Add Outfit

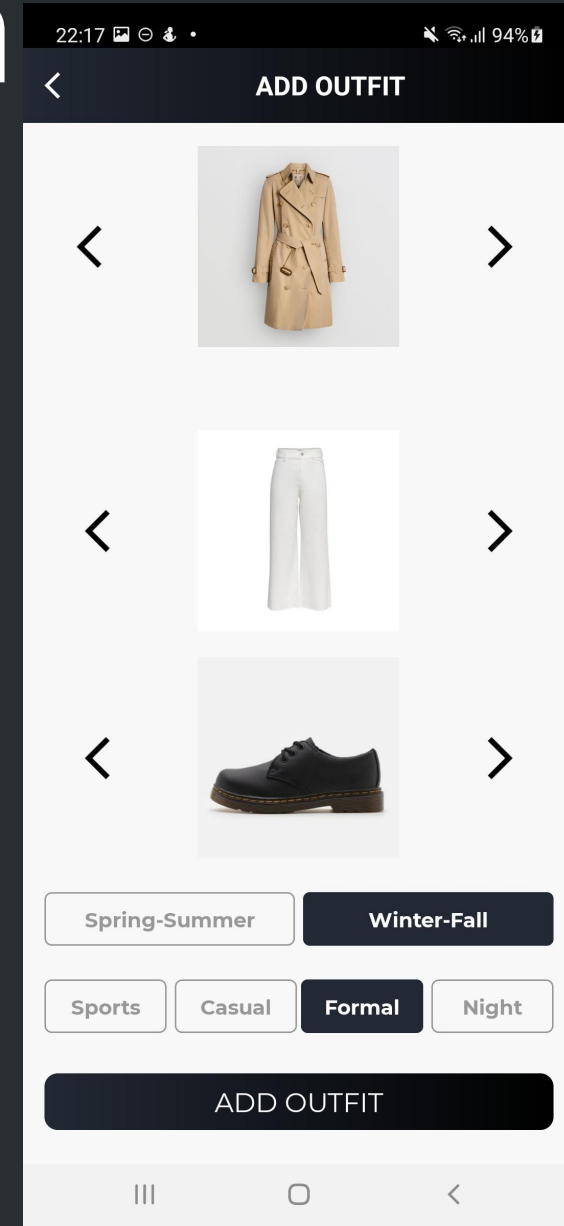


2. Bottom Navigation

CLOSET

DASHBOARD

OUTFIT → Add Outfit

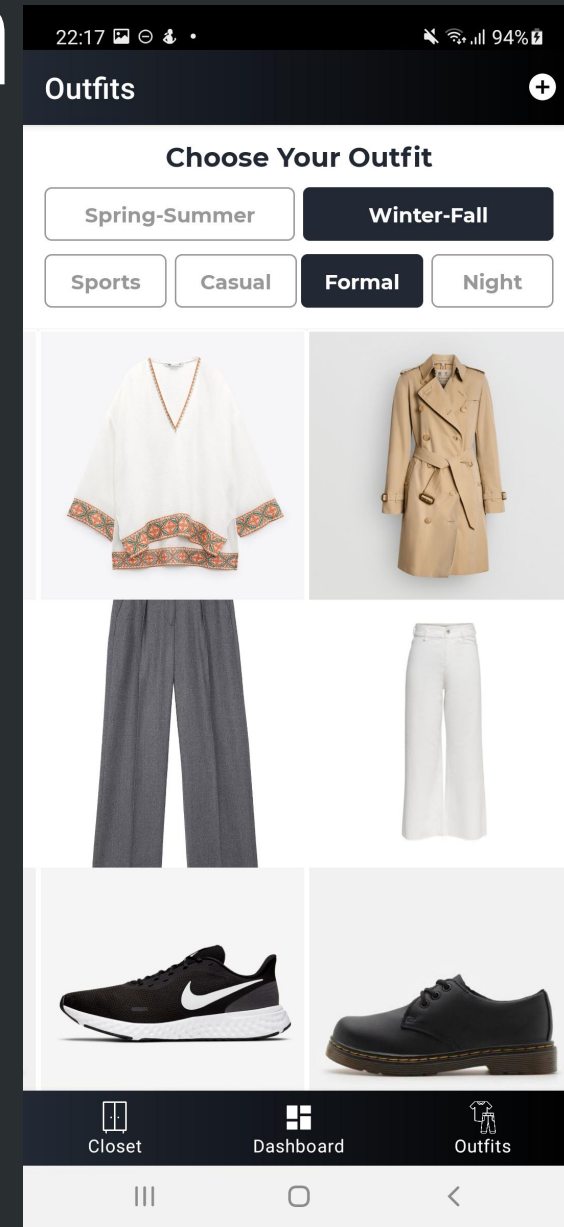


2. Bottom Navigation

CLOSET

DASHBOARD

OUTFIT → Add Outfit



3. Suggestion Algorithm



3. Suggestion Algorithm

- 1) Choose season either **WINTER-FALL** OR **SUMMER-SPRING**
- 2) Select the purpose **SPORTY CASUAL FORMAL NIGHT**
- 3) App will suggest an outfit that fits those descriptions and is clean (available) in closet
- 4) The user can ask the app to **retry** and make new suggestion or go back

4. Watch Implementation

Idea is to reduce cleanliness of outfit if the users heartbeat is above certain threshold
(i.e. if the user is running or hiking the items cleanliness go down quicker)

4. Watch Implementation

- 1) The watch is notified when we select a outfit and recover the cleanliness of the outfit.
- 2) The watch start to take the heart rate.
- 3) For a define interval of time, depending of the heart rate we reduce the cleanliness of the outfit.
- 4) The cleanliness of the clothes is shown on the watch.
- 5) The watch notified the smartphone of the new cleanliness of the clothes.

Thank you for your attention

Special thanks to our assistant,

Rafa Medina and professor Giovanni Ansaloni



DEMO